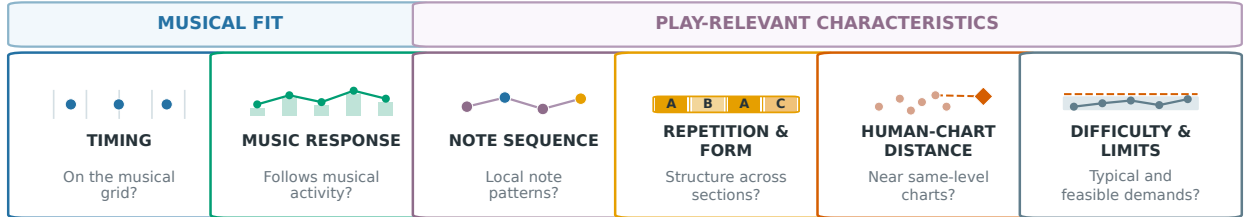


# ChartGenEval: A Multi-Dimensional Metric Suite for Rhythm-Game Chart Generation

Jhen-Ke Lin

National Yang Ming Chiao Tung University  
jacob.cs14@nycu.edu.tw



**Figure 1: The six dimensions of ChartGenEval.** ChartGenEval organizes chart evaluation into six distinct questions, grouped by musical fit and play-relevant characteristics. Each dimension retains its own interpretation.

## Abstract

A rhythm-game chart must fit its music and play well, yet one song admits many valid note sequences. Agreement with one official chart therefore conflates valid alternatives with errors. We introduce ChartGenEval, a six-dimensional metric suite that uses the matched official chart only for its authored timing map, never its note placements. For injected whole-chart shifts of 15, 30, and 60 ms, its median grid-phase estimates match the true shifts. By contrast, every primary chart-only output changes by less than 0.03 human standard deviations.

ChartGenEval reports this timing profile alongside five higher-level dimensions: note sequence, repetition and form, response to music, same-difficulty human-chart distance, and difficulty and playability limits. The dimensions remain separate, with no overall total. Nine nonredundant corruption-measurement pairs met prespecified sensitivity and invariance criteria on 80 held-out song groups; all 32 applicable control checks passed. The stress tests also show why isolated proxies can be misleading: common-pattern rewriting lowers language-model perplexity by 37%, while loop collapse raises self-similarity by 62%. Together, this automated six-dimensional profile can support fast comparison and iteration of chart-generation models. Depending on their roles, its outputs can serve as rewards, constraints, or monitoring signals in training methods such as reinforcement learning.

## 1 Introduction

Automatic chart generation for rhythm games maps audio to a playable sequence of timed, typed notes. Research has progressed from onset-detection methods to conditional sequence models [1–4]. Evaluation has not kept pace. A chart dictates what the player must do and when; it answers to

the music and to the player at once. We organize evaluation around two requirements. *Musical fit* asks whether notes share the song’s temporal structure and whether chart activity follows the music’s rise and fall. *Play-relevant characteristics* include learnable patterns, repetition with variation, and demands appropriate to the requested difficulty. A chart is a designed artifact: many note sequences can satisfy these requirements.

Existing methods answer useful but partial questions. The dominant evaluation, placement F1 against the one official chart, quantifies reconstruction: a valid alternative lowers overlap exactly as a flaw does, and no comparison is possible when the song has no reference chart. Optimizing agreement therefore rewards reproducing the official answer and penalizes the departures that make charting a design problem. Language-model (LM) likelihood quantifies model fit; distributional and self-similarity statistics describe generated material; and player or expert studies establish system-level experience. Each answers its own question, and an aggregate of them hides its trade-offs.

Free note choice still leaves a timing question. Conventional grid-based charts organize notes relative to a song’s musical-time framework. We measure this part of musical fit as alignment to an external grid. The matched official chart supplies its authored bar, meter, and tempo map, while its note placements are excluded as targets. This split preserves freedom of note choice while exposing chart-wide shifts that interval- and type-based statistics cannot observe. The proposed grid-phase offset gives the correct median estimate for injected shifts of 15, 30, and 60 ms, while every primary chart-only measurement remains nearly unchanged.

**ChartGenEval** automatically computes a profile across six dimensions: timing, note sequence, repetition and form, response to music, same-difficulty human-chart distance, and difficulty and playability. Its outputs serve three

roles: target signals for model improvement, feasibility constraints, and monitoring signals for model comparison and error analysis. These roles remain separate, so each dimension keeps its meaning.

Controlled corruptions provide the sensitivity evidence. Starting from a human chart, we apply a dose-controlled transformation designed to impair one property and ask whether its target measurement responds while the applicable control checks remain unchanged. The nine transformations cover dense timing jitter, sparse timing errors, a whole-chart shift, note-type shuffling, loop collapse, common-pattern rewriting, density scaling, burst insertion, and bar shuffling. We selected and revised candidate measurements on 40 development songs, then evaluated ten prespecified corruption–measurement pairs on 80 held-out song groups. The results meet all target and control criteria; one algebraically equivalent pair leaves nine nonredundant tests (Section 4.1). Development stress tests further show that LM perplexity can favor common-pattern rewriting, self-similarity can favor loop collapse, and chart-only measurements miss a constant chart–audio shift.

Our contributions are:

1. **Automated six-dimensional feedback.** ChartGenEval computes separate signals for timing, note sequence, repetition and form, response to music, human-chart distance, and difficulty and playability. These signals can support model comparison, iteration, and reward-based training.
2. **Controlled-corruption evidence.** Nine dose-controlled corruptions yield nine nonredundant held-out tests of target response and specified invariance checks. The same design exposes proxy sign reversals and the global-shift blind spot of chart-only measurements.
3. **Chart-specific measurements.** The suite adds a grid-phase offset and five development-stage components for chart–audio response, long-form structure, and human-chart distance.

## 2 Related Work

**Reference-chart agreement.** DDC and GenéLive! report F-scores for predicted placements, while TCP reports onset and onset-plus-type F1 together with local pattern recall [1, 3, 4]. Their tolerances and aggregation differ: DDC uses  $\pm 20$  ms, GenéLive! uses  $\pm 50$  ms, and TCP defines tolerances as fractions of a bar. TaikoNation instead compares 23 ms binary frames and pattern spaces [2]. These are reconstruction estimands and inherit the single-reference limits described in Section 1.

**Unpaired and human-anchored evaluation.** DDC uses LM perplexity and token accuracy to evaluate held-out model fit [1]. Symbolic-music and text-generation work contributes diversity, feature-distribution, self-similarity, and set-level repetition statistics [5–9]. Human studies answer another question. MuTE validates a reference-aligned piano-arrangement measure against expert ratings [10]; GenéLive! reports professional deployment feed-

back [3]; and Mania Archetype combines reference similarities with ratings from eight experienced players [11]. ChartGenEval supplies the automatic measurements and corruption-sensitivity evidence that such human studies can build on.

**Metric stress tests and timing.** SCHmUBERT constructs non-musical piano rolls that match framewise self-similarity statistics [12]. Fréchet Music Distance is tested with pitch and velocity perturbations, but remains a set-level embedding distance [13]. STRUM reports reference-onset F1 after a per-song global-offset search [14]; ChartGenEval reports the whole-chart grid-phase offset as a chart-level measurement without using reference notes as targets. Our 6 ms timing tolerance follows temporal discrimination research [15, 16] (Section 3.4).

## 3 Methods

### 3.1 ChartGenEval metric suite

ChartGenEval automatically computes six groups of measurements. Controlled corruptions test target sensitivity across doses, while applicable control checks test invariance. Table 1 distinguishes the measurements proposed here from adaptations of established chart statistics and states the strongest evidence available for each role.

All ChartGenEval outputs are computed automatically, but they serve different roles. Directional outputs can act as rewards or penalties, playability checks as constraints, and descriptive outputs as monitoring signals. A training method, including reinforcement learning, can choose and weight these signals for its task.

The proposed measurements target gaps left by individual chart statistics. The *whole-chart grid-phase offset* searches shifts from  $-250$  to  $+250$  ms against a fixed lattice of bar, beat, and eighth-note anchors. To test whether activity follows the music’s rise and fall, the *density–energy response* is the Spearman correlation between hits per second and linear mel energy in authored bar windows. It is unavailable with fewer than eight windows or when the song’s 90th–10th percentile energy range is below 5% of its median. The *run-head onset support* is the fraction of rhythmic-group heads whose maximum spectral flux within three mel frames ( $\approx \pm 35$  ms) reaches the song’s 60th-percentile threshold. A new group begins after an inter-onset interval of at least 0.25 s, preventing dense runs from dominating the denominator.

For long-form structure, *reciprocity* divides the chart into two-bar phrases on sixteenth-note slots. A tolerant, chance-corrected rhythm similarity is multiplied by chance-corrected color agreement. If  $S(i, j)$  denotes this product with one-slot tolerance, the score is the phrase mean of  $S(i, i + 2)[1 - S(i, i + 1)]$ : material returns after the intervening phrase without rewarding adjacent copy-paste repetition.

The *stagnation–alienation proxy* operates on active-bar rhythm and color fingerprints. A bar is stagnant when it is the fourth or later member of a consecutive similarity- $\geq .8$  run, or when every eligible 16-note color window is exactly periodic with period at most four. A bar is alienated when

| Construct             | Primary output and reading                                                           | Origin                                      | Evidence basis                                                             |
|-----------------------|--------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------|
| Timing                | Grid clean rate and residual tail; whole-chart grid-phase offset                     | Proposed chart-to-grid profile              | C1, C1s, C2 held-out tests using the authored grid                         |
| Note sequence         | LM loss as description; unseen-transition penalty as a one-sided familiarity check   | Adapted $n$ -gram statistics                | C3 held-out test (transition familiarity)                                  |
| Repetition and form   | 4-gram typicality; ABAB reciprocity; stagnation–alienation proxy                     | Adapted 4-gram score; two proposed measures | C4, C5, C8 held-out tests (one 4-gram axis); new measures development only |
| Response to music     | Bar-level density–energy rank correlation; run-head onset support                    | Two proposed measures                       | Development only                                                           |
| Human-chart distance  | Mean distance to 20 nearest training charts in a 32-dimensional chart representation | Proposed chart-specific instantiation       | Development only                                                           |
| Difficulty and limits | Density and strain typicality; overload and density-spike limits                     | Adapted statistics; proposed calibration    | C6, C7 held-out tests (density typicality, spike limit)                    |

Table 1: ChartGenEval constructs, origins, and evidence boundaries. Each construct’s required timing, audio, grid, or reference anchor is given in Methods (Section 3.2, Section 3.3). “Proposed” marks measurements defined in this work; their basic statistics may already be established. Held-out evidence refers only to the named outputs.

no bar with similarity at least .75 occurs within 32 bars in either direction. The raw output is the fraction of active bars in the union of these sets. We interpret the output as a structural signal. Finally, the *human-chart distance* represents each chart with 32 interval, note-type, density, and variety features. Features are standardized within course; the raw distance is the mean Euclidean distance to the 20 nearest training charts of that course. Its bounded display score is  $\exp(-g/a)$ , where  $g$  is the raw distance and  $a$  is the course’s 90th percentile leave-one-out distance. The representation is chart-specific; its entropy, bigram, and nearest-neighbor primitives are established statistics.

### 3.2 Data and timing references

We use taiko-style chart files and associated audio gathered from community resources available online. The chart files transcribe the official difficulty courses; we use them as human reference charts. Because the underlying charts and audio are potentially copyrighted content, the corpus itself is not distributed. Source rows are grouped by underlying song, with inconsistent source mappings rejected. The training and test splits serve different purposes. The training split has 924 source rows covering 813 unique song groups and 3,880 course charts. We use these charts to fit the calibration and LM. The test split has 140 source rows forming 120 song groups. The first 40 groups contain 170 human charts and form the development panel for metric and corruption design, figures, and system profiles. The remaining 80 groups contain 333 course charts and form the held-out panel for confirmatory tests. The dataset also has a validation split of 80 song groups, which is not used in the reported analyses. Across the development and held-out analyses, controlled corruptions cover all 120 test songs: 40 for development and 80 for held-out confirmation. The public-system comparison uses the 40 development songs.

The five additional proposed components were selected and revised on the development panel, so their evidence remains exploratory. Appendix A documents panel construction and the full analysis specification.

Timing requires an external song-timing reference. The primary grid comes from the matched official chart file’s

authored per-bar timestamps, meter, and tempo metadata. Its note times and note types are never used as targets: the official file supplies the timing map, not a single correct note sequence. This retains the map’s precise authored bar structure while allowing generated charts to make different valid note choices within the same musical-time framework. We never use the grid declared by the generated chart, because that would let the generator define its own reference.

We also evaluate audio-estimated downbeats as a development-stage alternative when an authored timing map is unavailable. On 39 of the 40 development songs, the estimated beat period is within 2% of the metadata-based period; the remaining song differs by 2.39%. Absolute-offset reporting uses the authored grid, which fixes bar phase directly from the timing map. Only courses with an authored timing map enter the held-out timing analysis; timing measurements are unavailable for the remaining courses.

### 3.3 Measurement roles and difficulty calibration

Each chart is a sequence of timed, typed notes. We organize the raw measurements around the six question groups in Figure 1, which divide between the two requirements of Section 1. Musical fit is agreement between what is heard and what is played, so timing and response to music require the song. The other four groups do not require audio and describe play characteristics; the proposed structure measurements additionally require a metrical grid. The adapted chart-side primitives are a smoothed trigram LM, unseen-transition rate, note rate, local density, short-interval load, entropy, and unique or repeated 4-grams. We retain trigram negative log-likelihood (NLL) as a description of LM fit; perplexity is its exponential. We transform NLL separately into a course-conditioned band score, termed *pattern-IC typicality*. Unseen-transition rate supplies a one-sided *transition-familiarity* check: an unseen trigram is unfamiliar to the training corpus, not semantically invalid. The proposed structure, audio-response, timing, and human-gap measurements are defined in Section 3.1.

We turn each selected feature into a *same-difficulty band score*. For every course and selected feature, let  $l$  and  $u$  be the 10th and 90th percentiles of the training charts and let  $h = \max((u - l)/2, 10^{-9})$  be the protected half-width. The distance is symmetric at the two edges:

$$d(x) = \begin{cases} (l - x)/h, & x < l, \\ 0, & l \leq x \leq u, \\ (x - u)/h, & x > u, \end{cases} \quad (1)$$

$$s(x) = \frac{1}{1 + d(x)^2}.$$

Thus a chart scores 1 inside the band, and equal departures below and above it receive equal scores. We also retain the signed distance so that large departures do not collapse to the same near-zero score.

This score measures how unusual a chart is for the requested difficulty; it does not by itself say whether the chart is good. Some measurements have a declared undesirable direction, such as excess load beyond a course-relative limit; these use separately named one-sided scores. Transition familiarity is also one-sided relative to the training corpus, without implying semantic invalidity. Family measurements without a declared calibration remain raw descriptions. Entropy, density, and repetition have no universal direction: each can be too low or too high. The controlled corruptions show that targeted changes often move these values away from the human band. An unusual chart that remains playable and follows the music may still be creative (Section 5).

Playability limits remain separate, non-compensable feasibility checks. On human charts in the development panel, 88%, 89%, and 97% score at least 0.99 on the note-rate, short-interval, and density-spike checks, respectively. Values such as estimated global shift remain descriptive.

### 3.4 Measuring timing against the song

We treat grid deviations below 6 ms as within tolerance, at the scale of the published just-noticeable difference for short intervals [15]. Larger errors are reported in three ranges: above 6, 12, and 18 ms. We also compare consecutive note intervals using a tolerance of  $\max(6 \text{ ms}, 2.5\% \times \text{IOI})$ . Every note is counted as within tolerance, above one of the three ranges, or unmatched. Unmatched generated notes remain in the denominator, so added off-grid notes lower the within-tolerance rate. The measure has no reference-note recall term: deleting unmatched generated notes can raise the rate. We therefore interpret it together with note count and same-difficulty density. We summarize each chart first, then report medians and tails across charts.

A common shift  $\delta$  leaves every inter-note interval unchanged:

$$(t_{i+1} + \delta) - (t_i + \delta) = t_{i+1} - t_i. \quad (2)$$

The note types are also unchanged. Chart-only measurements built from these intervals and types therefore stay unchanged or nearly unchanged.

Nearest-grid matching can miss a constant shift (Figure 2). On a dense grid, each shifted note may simply be paired with another nearby grid point. We therefore estimate one offset for the whole chart. Let  $G_k$  be the fixed bar, beat, and eighth-note grids, and let  $d(x, G_k) = \min_{g \in G_k} |x - g|$ . For note times  $t_i$ , we score a candidate shift by

$$S(\delta) = \sum_k w_k \frac{1}{n} \sum_i \left[ 1 - \frac{d(t_i - \delta, G_k)}{\tau} \right]_+, \quad (3)$$

$$\hat{\delta} = \arg \max_{\delta \in [-250, 250] \text{ ms}} S(\delta),$$

where  $[z]_+ = \max(z, 0)$ ,  $\tau = 12 \text{ ms}$ , and the bar, beat, and eighth-note weights are 0.2, 0.3, and 0.5. We search in 0.5 ms steps. The authored grid comes from the official chart’s timing map, not its note sequence; the audio-estimated grid comes from audio downbeats. Combining the three grids resolves most periodic ties; the smallest-magnitude offset is chosen for near-ties. Both timing summaries require one of these external grids. If neither an authored map nor an audio-estimated grid is available, timing is reported as unavailable. BPM alone supplies a period but no phase, so it cannot serve as a timing reference.

### 3.5 Controlled corruptions and acceptance rules

The development response matrix is descriptive. It reports standardized maximum-dose changes and the per-chart net direction: the fraction moving up minus the fraction moving down relative to each chart’s intact version. Appendix A gives the complete development calculation and the independent-LM check for C5.

We evaluated ten prespecified corruption–measurement rows on 80 held-out song groups. Five deterministic corruption replicates were averaged within chart and dose, and courses were averaged within song, leaving songs as the inferential unit. Each test used an oriented dose correlation and a maximum-dose-minus-sham contrast. We computed song-cluster bootstrap intervals and applied Bonferroni-adjusted 99.5% intervals across the ten rows. A response met the criterion when at least 72 songs were evaluable, every required control passed, and both adjusted upper bounds were below zero. Appendix A provides the full protocol.

The controls match the invariance claimed by each measurement family. They include exact re-evaluation, a joint shift of notes and grid, a Don–Ka color bijection for timing and density, and a whole-chart translation for chart-only measurements. A failed control prevents a target response from being counted as supported.

### 3.6 Evaluated systems

We evaluate five public generators spanning three note formats: Taiko note types, timing-only onset streams, and osu! position-based objects. Two systems produce Taiko note types: Mapperatorinator v32 [17], a community-developed, MIT-licensed framework for multi-gamemode osu! beatmap generation evaluated here in taiko mode, and the FDG-published TaikoNation system [2], run from

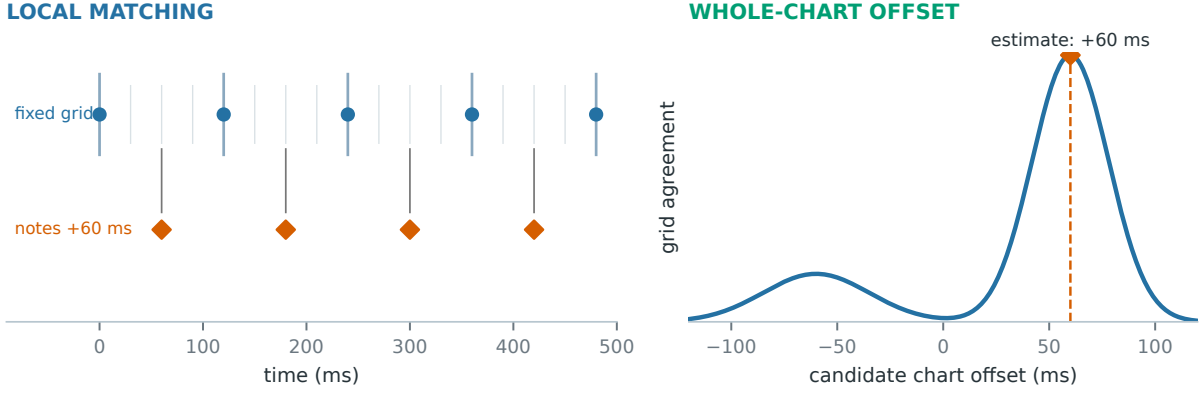


Figure 2: Why timing needs a whole-chart estimate. **Left:** nearest-grid matching can pair shifted notes with different subdivision points and report small local errors. **Right:** a schematic agreement curve over candidate whole-chart offsets; the curve peaks at the injected +60 ms shift. The grids are fixed before evaluating the generated chart.

| Controlled corruption (strength)                         | Changes               | Prespecified response                               |
|----------------------------------------------------------|-----------------------|-----------------------------------------------------|
| C1 timing jitter ( $\pm 10/20/30$ ms)                    | note timing           | timing clean rate $\downarrow$                      |
| C1s sparse timing errors (nominal 0.5/1/2%, $\pm 60$ ms) | a few note times      | absolute timing-error p99 $\uparrow$                |
| C2 global shift (+15/30/60 ms)                           | chart-audio alignment | absolute grid-phase offset $\uparrow$               |
| C3 note-type shuffle ( $p=0.2/0.4/0.8$ )                 | local transitions     | transition familiarity $\downarrow$                 |
| C4 loop collapse ( $p=0.3/0.6/1$ )                       | repetition and form   | 4-gram typicality $\downarrow$                      |
| C5 common-pattern rewrite ( $p=0.3/0.6/1$ )              | pattern variety       | 4-gram typicality $\downarrow$ (one effective axis) |
| C6 note-rate scale ( $\times 0.5/1.5/2$ )                | difficulty fit        | note-rate typicality $\downarrow$                   |
| C7 burst insert (2/4/8 clusters)                         | local overload        | density-spike limit $\downarrow$                    |
| C8 bar-order shuffle ( $p=0.3/0.6/1$ )                   | global form           | 4-gram typicality $\downarrow$                      |

Table 2: Controlled corruptions and their nine nonredundant target readings. C3–C5 preserve note times, while C4 and C6 preserve or control density. C5 originally named two algebraically equivalent 4-gram outputs; we count them as one effective axis. Pattern-IC typicality and raw trigram NLL remain direction-free secondary analyses. Appendix B provides visual before–after examples and the complete transformation specifications.

the authors’ public research repository. Three systems enter only the timing and music-response measurements: Dance Dance Convolution’s onset pipeline (ddconset) [1], the public timing-only configuration of GenéLive! [3], and AutoOsu [18], an osu! difficulty-targeted generator. All systems chart the same 40 development songs from audio in their public inference configurations. We did not train or fine-tune these systems on this corpus, although overlap between these commercial songs and the systems’ own training data cannot be excluded (Section 6). Course mappings are as follows. Mapperatorinator uses the corresponding Taiko course. GenéLive’s Beginner, Easy, Medium, Hard, and Challenge settings map to Easy, Normal, Hard, Oni, and Ura; when a song has no Ura chart, Challenge uses Oni as its reference course. Single-setting systems use the declared Oni reference. Format conversion into the evaluation event stream is shared within each note format (Section 4.5). Human charts of the same songs provide the reference row. Systems that condition on difficulty produce one chart per available course (170 or 200 charts); the others produce one per song (40). We report these profiles as descriptive development analyses.

## 4 Results

### 4.1 Nine effective tests meet the held-out criteria

All ten prespecified rows met both response criteria, all 32 applicable control checks passed, and every analysis retained all 80 song clusters (Table 3). The two C5 rows are algebraically equivalent, so the result represents nine nonredundant corruption–measurement pairs across seven effective output axes.

All 32 controls were exactly invariant. C1s retained 332 complete courses and C8 retained 330; the other pairs retained 333. Sample counts and exclusion reasons appear in Appendix A.

**C5 contributes one effective axis.** Repeated 4-gram rate is exactly one minus unique 4-gram rate. Their mirrored symmetric-band scores are therefore identical up to floating-point roundoff. The two prespecified C5 output names do not provide independent corroboration; C5 supports one effective 4-gram repetition/uniqueness response.

| Pair | Primary response           | Courses | $\hat{\rho}_{\text{dose}}$ [99.5% CI] | $\bar{\Delta}_{\text{max-sham}}$ [99.5% CI] |
|------|----------------------------|---------|---------------------------------------|---------------------------------------------|
| C1   | timing clean rate          | 333     | -.744 [-.815, -.667]                  | -.502 [-.558, -.446]                        |
| C1s  | absolute timing-error p99  | 332     | -.234 [-.344, -.134]                  | -.306 [-.508, -.144] ms                     |
| C2   | absolute grid-phase offset | 333     | -.916 [-.981, -.826]                  | -55.58 [-61.43, -49.83] ms                  |
| C3   | transition familiarity     | 333     | -.800 [-.856, -.737]                  | -.231 [-.262, -.199]                        |
| C4   | 4-gram typicality          | 333     | -.445 [-.573, -.312]                  | -.165 [-.226, -.108]                        |
| C5   | 4-gram typicality          | 333     | -.358 [-.459, -.258]                  | -.131 [-.189, -.078]                        |
| C6   | note-rate typicality       | 333     | -.693 [-.784, -.579]                  | -.582 [-.672, -.483]                        |
| C7   | density-spike limit        | 333     | -.940 [-.950, -.930]                  | -.619 [-.644, -.593]                        |
| C8   | 4-gram typicality          | 330     | -.589 [-.728, -.435]                  | -.224 [-.293, -.156]                        |

Table 3: Held-out results on 80 song groups. Larger oriented values are better, so target sensitivity appears as a negative dose correlation and a negative maximum-dose-minus-sham contrast. All nine effective pairs meet the prespecified criteria; intervals are family-wise 99.5% song-cluster bootstrap intervals. C5 reports one of two algebraically equivalent output rows.

## 4.2 External time detects chart–audio shifts

A chart can have plausible density, patterns, and note types while being late relative to the music. C2 shifts every note by 15, 30, or 60 ms. On the development panel, no primary chart-only output changes by more than 0.03 human standard deviations because interval- and type-based measurements are invariant to a common translation.

Nearest-grid matching also misses the shift: a dense lattice can pair each shifted note with a different nearby subdivision. The grid-phase offset search instead returns median estimates of 15.00, 30.00, and 60.00 ms from both reference sources. For every source and shift, at least 169 of 170 charts are within 3 ms of the injected value. Run-head onset support also responds at note onsets (net direction  $-0.74$  to  $-0.88$  across strengths) while remaining unchanged when note times are preserved. Under dense timing jitter, the fraction of notes within 6 ms falls in every chart at every strength; at  $\pm 10$  ms, the measured median outside the tolerance is 0.40, matching  $P(|U(-10, 10)| > 6) = 0.4$ .

Sparse errors require a tail-sensitive reading. Under C1s, nearest-grid matching re-pairs most moved notes and the cross-chart median of mean timing error remains zero. The timing-error p99 responds in held-out testing, while the note-sequence LM also changes because moving one note alters two adjacent intervals. Density remains essentially unchanged, and form responds only weakly.

## 4.3 Targeted corruptions expose favorable proxy movement

Two borrowed metrics move in the nominally favorable direction under a targeted corruption (Figure 3, left). *Perplexity favors common patterns*: human development charts average 9.44, while rewriting them toward the LM’s most common patterns lowers perplexity by 37% ( $9.44 \rightarrow 5.98$ ). Dance Dance Convolution reports perplexity as model fit on held-out human charts [1]; transplanted to generated output as a reference-free quality reading, a lower-is-better interpretation favors the rewritten chart. This is a chart-level example of the likelihood/quality dissociation [19]. The reversal remains when one LM creates the rewrite and another scores it ( $9.44 \rightarrow 6.04$ ,  $-36\%$ ). The pattern-IC band score moves toward its human band and increases across the three doses. In the held-out secondary analysis, its maximum-dose change is  $+0.021$  with a nominal 95% interval  $[-0.034, 0.077]$ , while raw trigram NLL falls

from 2.087 to 1.715 ( $-0.373$ ,  $[-0.399, -0.346]$ ). Typicality therefore does not repair this case by itself. The effective 4-gram axis (Section 4.1) falls from 0.99 to 0.87 in development.

*Self-similarity favors collapse*: replacing charts with their own dominant 4-gram loop raises self-similarity structureness [9] by 62% ( $0.0848 \rightarrow 0.1370$ ) and set-level Self-BLEU [6] from 0.845 to 0.898. Copy-paste output is more self-similar by construction, while the 4-gram repetition/uniqueness score falls from 0.99 to 0.83.

Two observational results complete the case. *Direction instability*: against the two publicly available taiko generators of Section 4.5, whose outputs lie on the high-entropy side of the human corpus, perplexity separates human from generated charts in the expected direction (AUC 0.80), yet C5 shows that the same reading inverts on the low-entropy side. A signed quality metric whose sign depends on which side of the human distribution the system sits cannot support a stable ranking; the same-difficulty band score treats both sides symmetrically as deviations from typicality. C5 shows why that score must be read together with the 4-gram repetition/uniqueness axis. *Length confounds*: on the same pool, type–token ratio correlates with note count at  $|\rho| = 0.85$  and distinct-2 at 0.56, against at most 0.28 for the band scores.

These cases establish complementarity without supplying aggregation weights. We therefore keep the outputs separate instead of hiding their trade-offs in one total. The complete metric–corruption response matrix appears in Appendix Figure 9.

## 4.4 Development evidence for the proposed components

The five additional components in Table 1 have descriptive development evidence. Their pooled Spearman responses across intact charts and three doses support distinct operational roles. Reciprocity falls under loop collapse and bar shuffling ( $\rho = -.612$  and  $-.506$ ) while remaining nearly unchanged under timing jitter and global translation ( $-.086$  and  $-.003$ ). The stagnation–alienation proxy responds strongly to loop collapse ( $\rho = -.924$  after orienting larger as better) and changes little under timing jitter or global translation ( $-.007$  and  $+.010$ ). These results support sensitivity to the targeted structural changes.

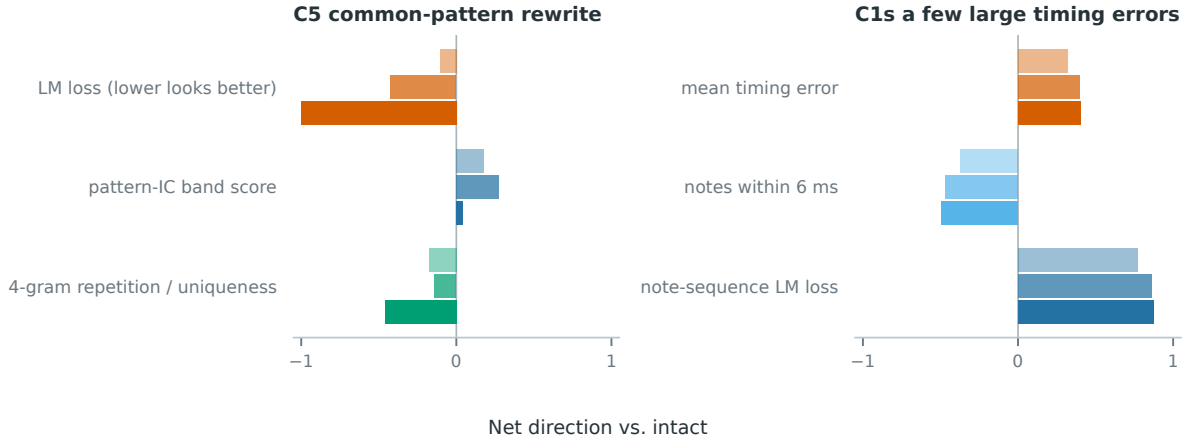


Figure 3: Two cases in which one measurement misses a targeted change and another responds. Bars show the fraction of charts moving up minus the fraction moving down relative to the intact chart; light to dark marks the three corruption strengths on the development panel (descriptive, not independent-chart confidence statements). **Left:** common-pattern rewriting lowers LM loss and raises the pattern-IC band score, while the 4-gram repetition/uniqueness axis moves the other way (its two algebraically equivalent output names are plotted once). **Right:** sparse  $\pm 60$  ms timing errors leave the mean timing error near zero and are mostly re-paired by the nearest grid, while the note-sequence LM responds to the changed intervals.

The two audio-response measures capture different scales. Run-head onset support decreases under a global shift ( $\rho = -.559$ ), whereas bar-level density–energy response is stable under uniform density scaling ( $\rho = +.005$ ) and falls under bar shuffling ( $\rho = -.386$ ). Its weak burst-insertion response ( $\rho = -.092$ ) confines the reading to bar-scale covariation. The same-course human-chart distance describes a joint feature distribution: it flags combinations unlike the training cloud without classifying unusual charts as creative or poor. Its course-conditioned score has maximum-dose net directions from  $-.32$  to  $-.88$  under C3–C8 and a median change of zero under global translation.

## 4.5 System profiles

The descriptive system profiles in Figure 4 and Figure 5 separate two effects that a single number would merge: similar headline timing fractions can arise from different timing behavior, and the requested course can change the reading of chart-side measurements.

**Similar timing fractions can have different causes.** In Figure 4, the near-unity human row is a consistency check on using the authored bar metadata as the reference, not a model result. The two Taiko systems place nearly the same median fraction of notes within 6 ms (0.59 and 0.58). Their remaining timing behavior differs. Mapperatorinator places 10% of adjacent intervals outside the relative tolerance, compared with 41% for TaikoNation. The grid-phase offset search finds median offsets of 26 and 55 ms, while nearest-grid matching reports less than 1 ms mean error beyond the 6 ms tolerance for both. TaikoNation also shows almost no relation between chart activity and audio energy (correlation  $-0.003$ ), and its notes fall on average-energy frames. The DDC onset system gives a 2.5 ms offset on the same audio, serving as a control against a shared decoding shift. AutoOsu, which shares Mapperatorinator’s osu!

conversion path, reads a 19 ms median offset; the differing values rule out a shared conversion constant as the sole explanation (Section 3.6).

**The requested course changes the reading.** Mapperatorinator’s generated-to-human note-rate ratio falls from 2.15–2.35 on Easy and Normal to 1.31 on Oni and 1.07 on Ura (Appendix Figure 10). Its note-rate, short-interval, and local-pattern scores are below 0.5 on Easy through Hard and near 1 on Oni and Ura. Timing does not follow the same trend: median fractions within 6 ms are 0.56, 0.60, 0.59, 0.57, and 0.62. The supported claim is therefore a difficulty mismatch, not a general improvement at high difficulty. TaikoNation exposes only one setting; at that setting, note rate and short-interval load fit the human range, while 55% of its trigrams are unseen in the course-conditioned training LM and its repeated-4-gram rate is 0.004.

## 5 Discussion

ChartGenEval provides automatic six-dimensional feedback for chart generators. The held-out tests show that seven effective output axes respond to their targeted corruptions, while the development analyses expose blind spots in common proxies. The profile can compare model versions and show which dimension changed. Directional outputs can act as rewards or penalties, playability checks as constraints, and descriptive outputs as monitoring signals. A reinforcement-learning setup can choose and weight these signals for its task.

**Atypicality requires context.** Controlled corruptions show that targeted changes often move scores outside the same-course human band, but the converse does not follow: an unusual chart may represent a creative choice. Timing and feasibility checks help interpret that departure. When

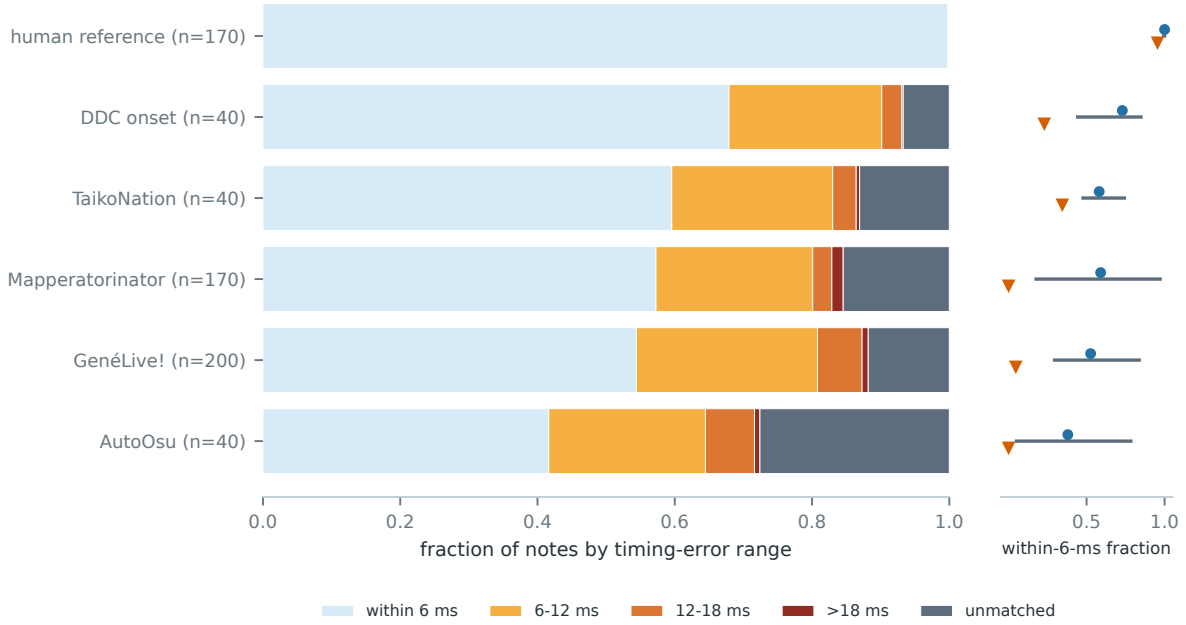


Figure 4: Timing of the evaluated systems on the 40 development songs using the authored grid. Each bar is the mean fraction of notes within 6 ms and in the 6–12, 12–18, and above-18 ms ranges, or unmatched;  $n$  is the number of charts. In the right strip, the horizontal line spans the 10th–90th percentiles, the dot marks the median, and a downward triangle marks the minimum within-6 ms fraction. Similar mean clean fractions can coexist with different timing tails.

|                  | Timing<br>within 6 ms | Note<br>rate | Short<br>intervals | Familiar<br>transitions | Local<br>patterns | 4-gram<br>repetition /<br>uniqueness | overload | density<br>spike | playability |
|------------------|-----------------------|--------------|--------------------|-------------------------|-------------------|--------------------------------------|----------|------------------|-------------|
| human reference  | 1.00                  | 1.00         | 1.00               | 0.75                    | 1.00              | 1.00                                 | ✓        | ✓                | ✓           |
| Mapperatorinator | 0.59                  | 0.23         | 0.36               | 0.23                    | 0.19              | 1.00                                 | ✓        | ✓                | ✗           |
| TaikoNation      | 0.58                  | 1.00         | 1.00               | 0.00                    | 0.01              | 0.08                                 | ✓        | ✓                | ✓           |

Figure 5: Separate development results for the human reference and the two generators that use Taiko note types. The first column is the median fraction of notes within 6 ms of the grid; the other cells are calibrated scores in  $[0, 1]$ . For note rate, short intervals, local patterns, and the 4-gram repetition/uniqueness score, 1 means the median lies within the human band; familiar transitions is one-sided (larger better), and even the human reference reads 0.75 because an LM still meets unseen trigrams. Symbols on the right show the overload, spike, and playability checks at a median threshold of 0.5. Other note formats appear only in the timing comparison (Figure 4). The separate columns expose different departures; they are not combined into a total.

signals disagree, the profile keeps the trade-off visible. The distance scale puts large departures in context: the development human maximum is 4.8 half-range widths on the sequence axis, while the Mapperatorinator and TaikoNation medians are 2.1 and 12.

**Transfer follows measurement roles.** Timing and music-response measurements read only note times, which allowed direct application to three note formats. Sequence, form, and distance measurements require a tokenizer for

the target game and human charts for calibration. Lane, position, or direction tokens can replace Taiko note types, and the same controlled-corruption design can test which responses transfer. Position-based games additionally require movement-cost, hand-allocation, and rotation checks.

## 6 Limitations

**Evidence scope.** Controlled corruptions show that measurements respond to known changes; they do not directly measure player experience. The nine corruptions omit wrong-song pairing, deletion of musically important notes, and deliberate off-grid placement. Individual charts may also move toward a human band after a corruption. The five additional components have only development-panel evidence. Finally, the two prespecified C5 output names are algebraically equivalent; future work should test an independent structure measurement.

**Corpus and system scope.** One game corpus defines the human ranges. Its difficulty and tempo coverage is summarized in Appendix Figure 6. Other note formats receive only timing and music-response measurements, and each public system runs in one public configuration at one seed. Possible overlap between development songs and system training corpora remains uncontrolled. The shared-path comparison argues against one conversion constant as the sole source of the observed offsets, but format-specific conversion effects remain possible. The system results are therefore comparative descriptions.

**Measurement and inference.** Timing clean rate penalizes added off-grid notes but has no reference recall term, so deletion can improve it; the measurement also requires an external grid. Held-out timing results use the authored grid, while the audio-estimated alternative has development evidence only. Sequence tokens include note intervals, so timing and sequence responses can overlap (Appendix Figure 9). Courses are averaged within song, leaving songs as the inferential unit. Deterministic corruption replicates are averaged within chart and do not create additional samples.

## 7 Conclusion

We introduced ChartGenEval, a six-dimensional metric suite that uses the matched official chart as an authored timing map while excluding its note placements as targets. Timing is reported alongside five higher-level dimensions; raw descriptions, two-sided typicality, and one-sided feasibility checks retain separate roles, with no overall score.

Nine nonredundant corruption–measurement pairs met the held-out sensitivity and invariance criteria on 80 song groups. The same study shows that chart-only measurements miss a global shift, while perplexity and self-similarity can move favorably under targeted corruptions.

The resulting profile provides automatic feedback for comparing and iterating chart generators. Directional outputs can guide reward-based optimization, feasibility checks can act as constraints, and descriptive outputs can track model behavior. This role-based design can support reinforcement learning while keeping each dimension separate.

## Ethics Statement

The study uses community-transcribed charts and associated audio gathered from online sources as described in Section 3.2; these potentially copyrighted materials are not

redistributed. The released materials exclude source charts, audio, personal data, and chart generators. More capable evaluation can lower the cost of automated content production; platform policy and moderation therefore remain necessary for high-volume deployment.

## AI Usage Statement

AI assistants were used for experiment orchestration, drafting, and editing under author direction; the author checked all reported values.

## A Corpus Coverage and Held-Out Evaluation Protocol

### A.1 Corpus coverage

The calibration corpus is summarized by course, star level, and tempo. Figure 6 shows its difficulty and tempo coverage.

### A.2 Panel construction and integrity

Source rows are canonicalized by `group_id` and audio SHA-256; inconsistent aliases are rejected before panel indexing. The test split contains 120 canonical song groups in stored order. Groups 0–39 form the development panel, and groups 40–119 form the held-out panel. Before inspecting held-out results, we fixed the dataset revision, corruptions and doses, target measurements, controls, thresholds, calibration, and analysis plan.

### A.3 Replicates, completeness, and inferential units

Every nonzero corruption cell and control condition uses five fixed base seeds. Replicates are averaged within chart and dose. A course is complete only when its intact value, every target dose and replicate, specified sham, and required controls are available. A song enters a test with at least one complete course, and complete courses are averaged within song. Songs, rather than charts, courses, or replicates, are the inferential units for both response statistics. Deterministic no-op and missing cells do not enter the analysis.

### A.4 Control checks and numerical invariance

Identity re-evaluation applies to every retained measurement. A joint +1-second shift of notes, authored grid, and duration tests invariance to the time origin. A Don–Ka color bijection applies to timing and density, where color is irrelevant; it is not a grammar or form control. The 60 ms C2 translation is the sham for chart-only density, grammar, and structure measurements. A required control failure prevents support for the affected target response. Each measurement uses a specified equivalence tolerance and timestamp-rounding rule. Chart-only interval arithmetic uses integer microseconds, three orders of magnitude finer than the smallest reported tolerance.

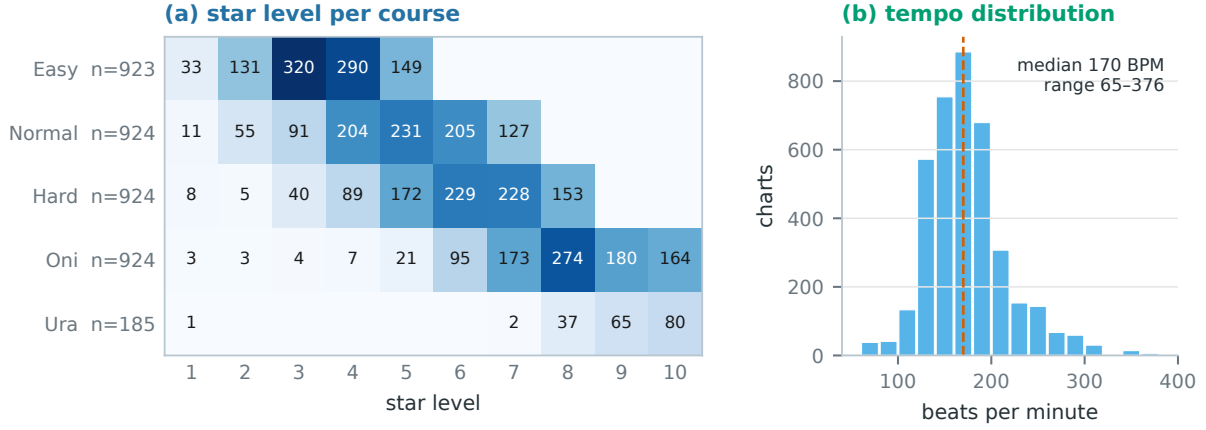


Figure 6: Course difficulty and tempo coverage of the human reference corpus that defines the calibration bands. **(a)** Star-level counts per course across the 3,880 training charts: each course spans a broad, shifting difficulty range. **(b)** Tempo histogram (median 170 BPM, range 65–376).

### A.5 Intervals and decision rule

After orienting outputs so that larger is better, the first statistic is the within-chart Spearman correlation between the four severities (intact plus three doses) and the measurement; an all-constant response has zero sensitivity. The second statistic is the oriented maximum-dose value minus the specified sham. Point estimates average song-level values. We use 10,000 song-cluster bootstrap draws: secondary analyses report nominal 95% intervals, and the ten primary rows use Bonferroni-adjusted 99.5% intervals. A response is supported when at least 72 of 80 songs are evaluable, all controls pass, and both adjusted upper bounds are below zero. A nonnegative adjusted lower bound contradicts the response; other cases are inconclusive. The two C5 output rows were required to meet the rule together, although their algebraic equivalence leaves one effective axis. The C5 LM uses the full stored-order training split, trigram order 3, and  $\text{add-}\alpha = 0.05$ .

### A.6 Sample counts and exclusions

The held-out execution produced 50,283 observation records from 333 charts. C1s retained 332 complete courses and C8 retained 330; every other effective pair retained 333. All 32 controls passed with maximum observed absolute difference zero. Nineteen deterministic no-op cells were excluded.

## B Controlled-Corruption Specifications

Figure 7 visualizes the operational change made by each corruption and its prespecified target measurement.

The 170-chart development matrix uses one realization per nonzero cell, producing 4,760 variants including the originals. C1s moves  $\max(1, \text{round}(np))$  of a chart’s  $n$  notes by  $\pm 60$  ms. C3–C5 preserve note times, while C4 and C6 preserve or control density. Raw features and their derived scores are evaluated on the same variants; equiva-

lent quantities are counted once. For C5, one LM chooses common-pattern rewrites and an independently fitted LM scores them. The perplexity reversal remains ( $9.44 \rightarrow 6.04$ ,  $-36\%$ ).

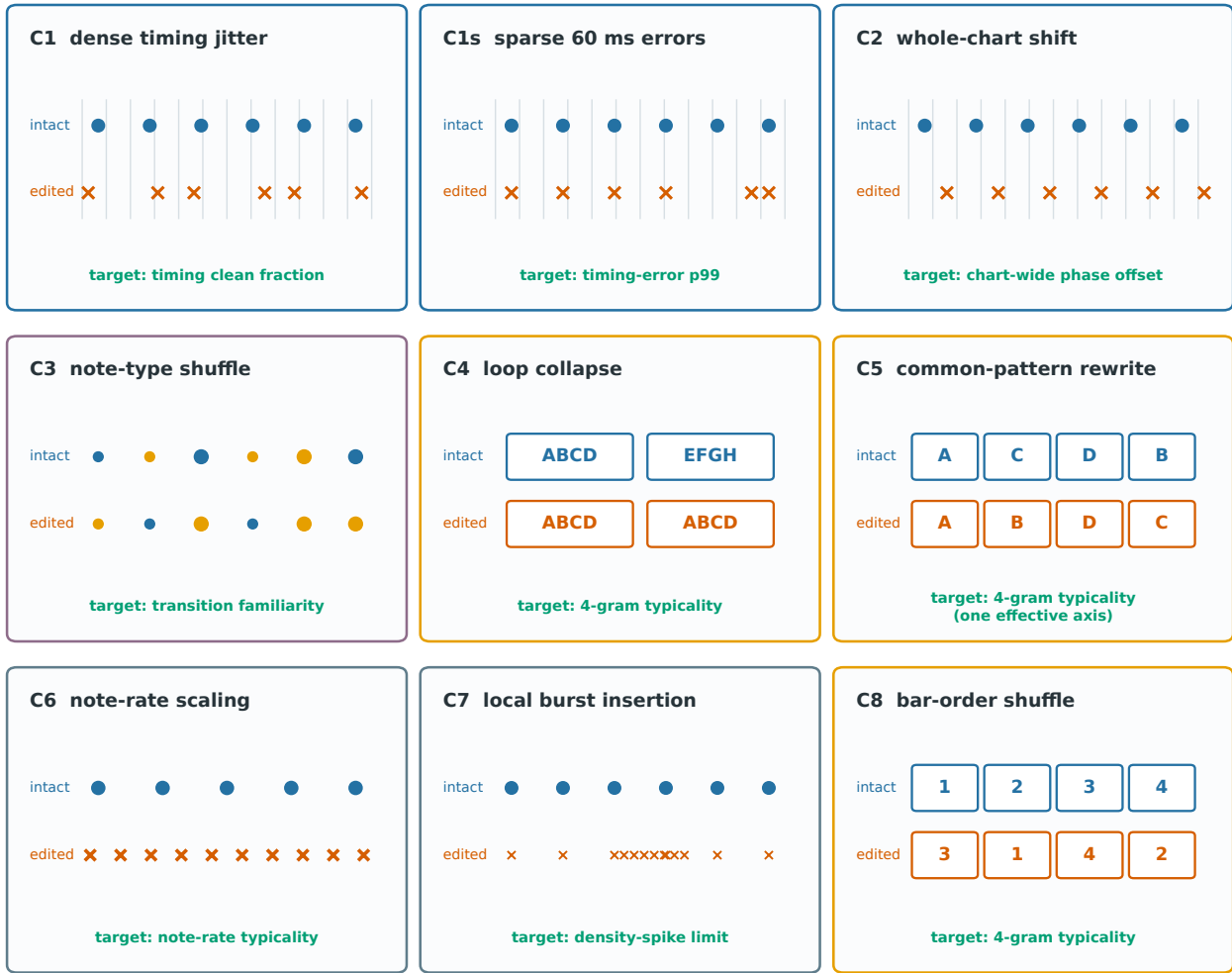


Figure 7: Controlled-corruption atlas. Blue marks the intact chart and orange marks edited events. Each panel shows the property changed by one corruption and names the target measurement. C1–C2 target timing, C3 targets transition familiarity, C4–C5 and C8 target repetition or form, and C6–C7 target difficulty fit and local load. These edits isolate operational changes; they do not exhaust chart quality.

## C Metric Guide and Extended Results

Figure 8 connects each construct to its required inputs, representative outputs, and strongest present evidence. The evidence labels distinguish outputs with held-out corruption tests from components supported by development analyses.

Figure 9 shows that targeted corruptions can affect multiple score families, motivating interpretation as a profile rather than as isolated interchangeable measurements.

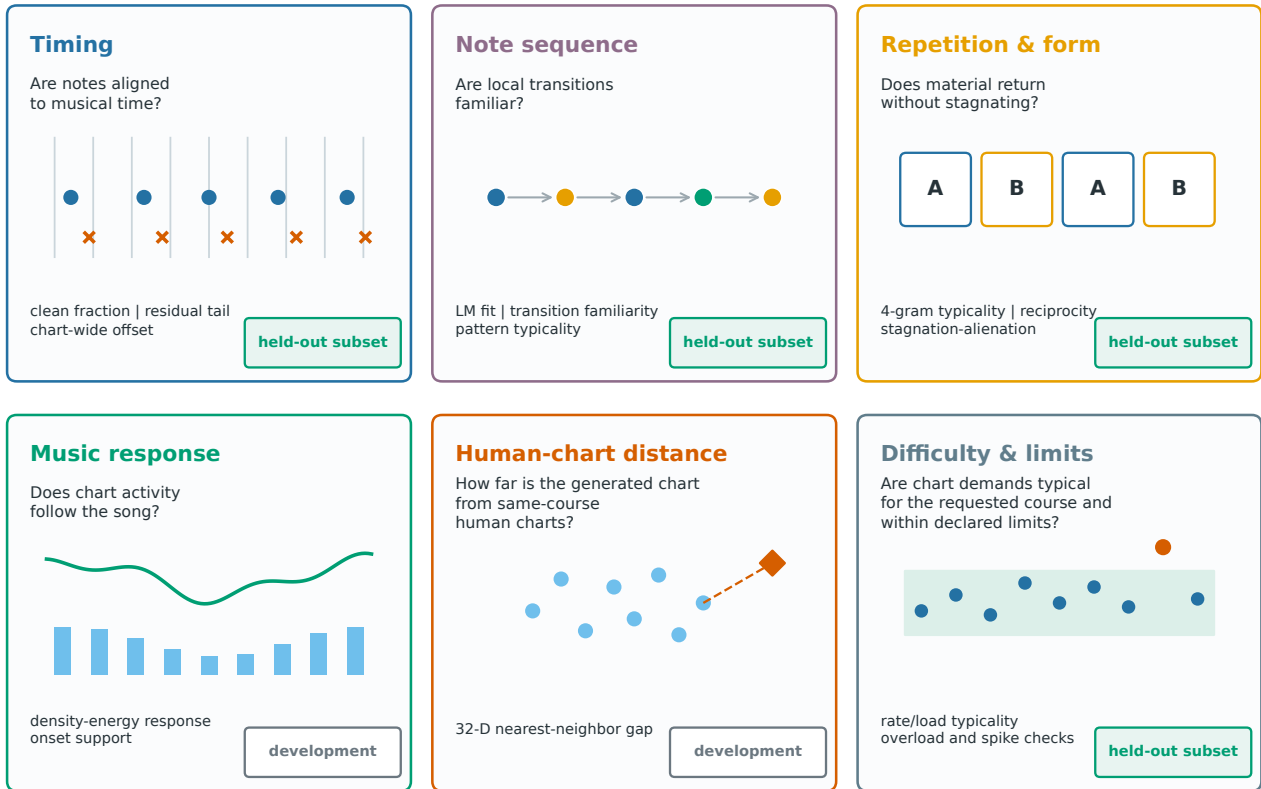


Figure 8: Metric-to-construct guide. Each panel states an evaluation question, illustrates the relevant signal, lists representative outputs, and marks the strongest evidence. “Held-out subset” means that only the named outputs in Table 1 have held-out corruption evidence. Typicality describes departure from same-course human ranges.

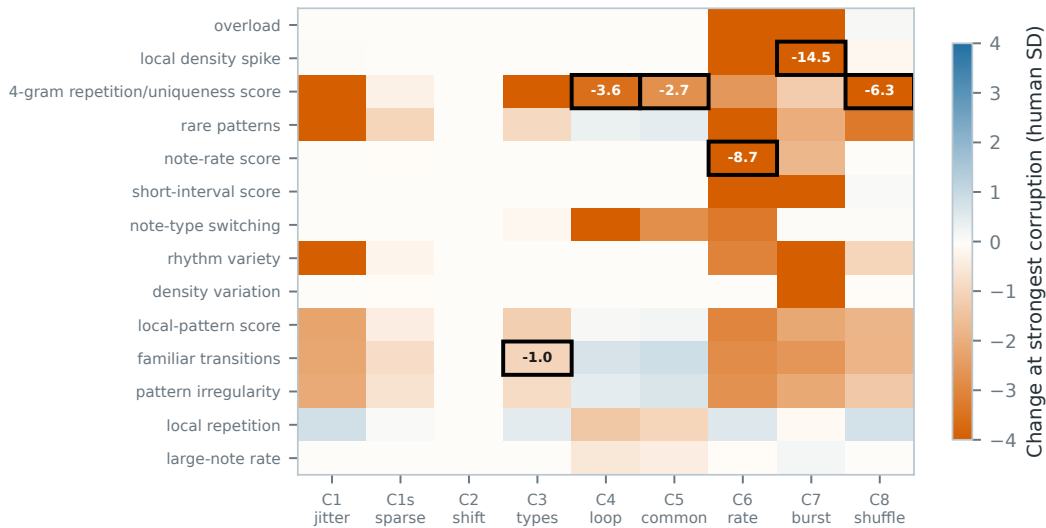


Figure 9: How 14 nonredundant calibrated chart-only scores respond to each corruption at maximum strength on the development panel. Cells cover band-typicality scores and one-sided checks. Red means the score falls and blue that it rises; values are in standard deviations of the human reference charts, clipped at  $\pm 4$ . Black outlines mark the target rows selected on the development panel; they do not mark statistical significance. The matrix shows both target sensitivity and cross-family coupling.

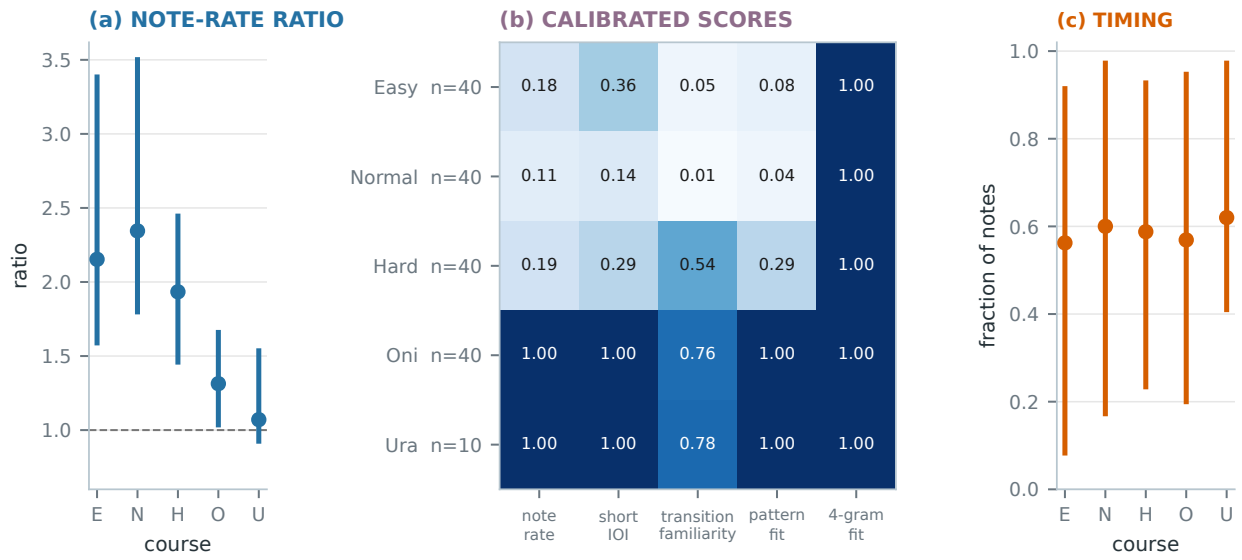


Figure 10: Exploratory difficulty profile for Mapperatorinator on paired development charts. Easy through Oni contain 40 song–course pairs; Ura contains 10. Dots are medians and vertical lines span the 10th to 90th percentiles. Generated density approaches the human reference as course difficulty rises, while timing does not follow the same trend; this supports a course mismatch rather than general improvement at high difficulty.

## D Released Materials

Code, evaluation records, and plotting scripts are available at <https://github.com/JacobLinCool/ChartGenEval>.

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